

SIFAT SADAKIN

+880-1988058487 | [LinkedIn](#) | [LeetCode](#)
sifat.sadakin@g.bracu.ac.bd | Dhaka, Bangladesh

EDUCATION

Bachelor of Science in Computer Science

Jan. 2023 – Jan. 2027

BRAC University; CGPA: 3.72/4.00; Expected Graduation: Jan. 2027

Dhaka, Bangladesh

Relevant Coursework: Data Structures, Algorithms, Artificial Intelligence, Database Systems, Software Engineering, Operating Systems, Statistics & Probability, Linear Algebra.

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Dec. 2025 – Present

BRAC University

Dhaka, Bangladesh

Efficient Self-Filtered PatchCore for Unsupervised Fabric Defect Detection: Memory-Bank Compression, Cross-Dataset Validation, and Edge Deployment

- Work under Senior Lecturer [Shahed Alam](#) on unsupervised fabric defect detection using PatchCore and deep visual features, with reproducible PyTorch training, evaluation, and ablation pipelines.
- Study memory-bank compression, self-filtering, cross-dataset generalization, inference latency, and edge-deployment tradeoffs while contributing to literature review and manuscript preparation.

Thesis Research

2026 – Present

BRAC University

Dhaka, Bangladesh

Robust Real-vs-AI Image Detection and Generator Attribution via Self-Supervised Representation Learning, Latent-Space Analysis, and Explainable AI

- Investigate representations for real-vs-AI image detection and generator attribution across unseen generators and image transformations.
- Use explainable AI and representation analysis to interpret model decisions, visualize class separability, analyze failure cases, and evaluate robustness to compression, resizing, and post-processing.

PROJECTS

Relay | AI Support Ticket Assistant

2026

Python, FastAPI, scikit-learn, SQL

- Built a support-ticket assistant that predicts ticket category and priority, checks similar past issues, and retrieves useful answers from a small knowledge base.
- Built a lightweight API and evaluation workflow so the model can be tested as a complete application rather than only in a notebook.

Sentinel | Transaction Risk Scoring

2026

Python, Pandas, scikit-learn, FastAPI

- Built a tabular ML pipeline that preprocesses transaction data, handles class imbalance, and returns a risk probability for each transaction.
- Exposed predictions through an API and evaluated the model using precision-recall metrics and feature-importance analysis.

Mosaic | Personalized Recommendation System

2026

Python, Pandas, scikit-learn, FastAPI

- Built a recommendation pipeline using user-item interactions and content similarity, including a practical fallback for new users with limited history.
- Evaluated ranking quality and served configurable top-N recommendations through a lightweight API.

Signal | Log Anomaly Detection

Python, Pandas, scikit-learn, SQL

- Converted application logs into frequency and error-rate features and used Isolation Forest to flag unusual activity.
- Grouped nearby alerts into incidents and stored results in a simple database workflow for easier review and analysis.

2026

Horizon | Demand Forecasting

Python, Pandas, scikit-learn

- Built a forecasting pipeline using lag features, rolling statistics, and time-aware train-test splits to predict future demand.
- Tracked MAE and MAPE across experiments to compare model changes and understand forecasting performance over time.

2026

TEACHING & LEADERSHIP EXPERIENCE

Undergraduate Teaching Assistant

Office of Academic Advising (OAA), BRAC University

- Support CSE110 through weekly in-person lectures and tutorials on programming fundamentals, along with consultation hours and one-to-one or group follow-up support.
- Monitor academic progress and help students develop structured study habits, programming fundamentals, and problem-solving strategies.

Jun. 2025 – Present
Dhaka, Bangladesh

Student Mentor | First Year Advising Team (FYAT)

BRAC University

- Mentor 100+ first-year students across multiple intakes through in-person and online academic and transition guidance.
- Received the Outstanding Mentor Award four times for sustained student support and mentorship.

Jan. 2025 – Present
Dhaka, Bangladesh

TECHNICAL SKILLS

Programming: Python, Java, SQL, C, C++

AI / Machine Learning: PyTorch, scikit-learn, OpenCV, TensorFlow/Keras (basic), NumPy, Pandas, computer vision, anomaly detection, explainable AI, model evaluation

Software / Backend: FastAPI, REST APIs, PostgreSQL, Django (basic)

Developer Tools: Git/GitHub, Docker (basic), Linux, VS Code, LaTeX

HONORS & AWARDS

- Outstanding Mentor Award (4x) - First Year Advising Team, BRAC University.
- Dean’s List and Vice Chancellor’s List - BRAC University.
- Merit-Based Scholarship Recipient - BRAC University.